

Indices - The Zero Index (Sheet 1)

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Use your calculator to find these values.

Example $5^3 = 125$

Calc: $5 \times \boxed{5} \times \boxed{5} =$

1 $7^3 =$ 2 $7^2 =$

3 $7^1 =$ 4 $7^0 =$

5 $9^3 =$ 6 $9^2 =$

7 $9^1 =$ 8 $9^0 =$

9 $4^3 =$ 10 $4^2 =$

11 $4^1 =$ 12 $4^0 =$

13 Look at Q.4, Q.8 and Q.12

The answers were all



Now use your calculator to complete these divisions.

14 $7^4 \div 1 =$

Calc: $7 \times \boxed{7} \times \boxed{7} \times \boxed{7} \div \boxed{1} =$

15 $7^4 \div 7^0 =$

$\therefore 7^0 =$

16 $10^3 \div 1 =$

17 $10^3 \div 10^0 =$

$\therefore 10^0 =$

Now use your division skills with index laws. No calculator

18 $3^5 \div 3^0 = 3^{\boxed{}} - \boxed{} = 3^{\boxed{}}$

19 $5^7 \div 5^0 = \boxed{}^{\boxed{}} - \boxed{}^{\boxed{}} = \boxed{}^{\boxed{}}$

When you divide a number (not 0) by '1', the number is also the answer.

$\therefore 3^0 = 5^0 =$

Answer these divisions in index form, then as numbers.

Example

Index Form

Number

$7^4 \div 7^4 = 7^0 = 1$

20 $5^2 \div 5^2 = \boxed{}^{\boxed{}} = \boxed{}$

21 $4^8 \div 4^8 = \boxed{} = \boxed{}$

22 $\frac{10^3}{10^3} = \boxed{} = \boxed{}$

23 $14^5 \div 14^5 = \boxed{} = \boxed{}$

24 $\frac{9^{11}}{9^{11}} = \boxed{} = \boxed{}$

Algebra is answered the same way.

25 $x^7 \div x^7 = \boxed{}^{\boxed{}} = \boxed{}$

26 $\frac{m^4}{m^4} = \boxed{} = \boxed{}$

27 $y^8 \div y^8 = \boxed{} = \boxed{}$

28 $\frac{n^{13}}{n^{13}} = \boxed{} = \boxed{}$

Multiply these answering in index form, then as numbers.

Example

Index Form

Number

$5^3 \times 5^0 = 5^3 = 125$

Calc: $5 \times \boxed{5} \times \boxed{5} =$

29 $4^6 \times 4^0 = \boxed{}^{\boxed{}} = \boxed{}$

30 $2^9 \times 2^0 = \boxed{} = \boxed{}$

31 $3^7 \times 3^0 = \boxed{} = \boxed{}$

Algebra same, but leave in index form

32 $v^7 \times v^0 = \boxed{}^{\boxed{}}$

33 $k^9 \times k^0 = \boxed{}$

34 $n^3 \times n^0 = \boxed{}$



Remember with brackets you multiply the powers. Expand these.

Example

Index Form

Number

$(7^3)^0 = 7^0 = 1$

35 $(5^6)^0 = \boxed{}^{\boxed{}} = \boxed{}$

36 $(2^4)^0 = \boxed{} = \boxed{}$

37 $(3^0)^6 = \boxed{} = \boxed{}$

38 $(x^0)^6 = \boxed{} = \boxed{}$

39 $(a^0)^3 = \boxed{} = \boxed{}$

40 $(q^5)^0 = \boxed{} = \boxed{}$



These have co-efficient and so are a little harder.

Example

$x^0 = 1$

$(3x^0)^3 = (3 \times 1)^3 = (3)^3$

Calc: $3 \times \boxed{3} \times \boxed{3} = 27$

41 $(5n^0)^4 = (\boxed{} \times \boxed{})^4 = \boxed{}^4 = \boxed{}$

42 $(8p^0)^3 = (\boxed{} \times \boxed{})^3 = \boxed{}^3 = \boxed{}$

43 $(2h^0)^9 = (\boxed{} \times \boxed{})^9 = \boxed{}^9 = \boxed{}$

When the '0' is outside, answer is 1

44 $(2h^3)^0 = \boxed{}$

45 $(8k^5)^0 = \boxed{}$

46 $(a^2b^5c^3)^0 = \boxed{}$